

**Research Methods in Biomedical Informatics**  
**Programming Assignment**  
**Spring 2026**

## Overview

This repo contains code that finds the longest contiguous sequence that contains a query sequence from a query sequence and fasta format file of next-gen sequencing reads.

*Find the assignment description here. Find the assignment report here.*

## Set up

Set up your conda environment using the environment file `environment.yaml`

```
conda env create -f environment.yaml`
```

## Source Descriptions

### `main.py`

- Takes next-generation sequencing reads identified in a sample and an initial query sequence and returns the largest sequence contig that can be constructed from the reads that contains the initial query sequence

### `functions.py`

- Contains helper functions
  - `load_data`
    - \* reads fasta files into a dictionary
  - `parse`
    - \* separates fasta sequences and headers of a list of fasta lines
  - `segment`
    - \* count each kmer
  - `segment_all`
    - \* wrapper to apply segment to all sequences in a list
  - `summarize`

- \* calculate stats (e.g. mean, sd) of an array
- `seq_summary`
  - \* `summarize` string lengths
- `align`
  - \* check if two strings match
- `make_adj`
  - \* make adjacency matrix between every combination of strings in Counter where matches fill matrix with a 1, and non-matches a 0
- `make_contigs`
  - \* take an adjacency matrix from `make_adj` and recursively following all possible paths through it, appending contigs along the way

#### `simulate_reads.py`

- Simulates random reads of a specific length and writes these to a fasta format including the ground truth sequence

## Execution

Example execution for next-gen reads `reads.fasta` and a query sequence `query.fasta` both in fasta format.

```
working_directory/
- in/
  - reads.fasta
  - query.fasta
- out/

cd working_directory
./main.py --reads in/reads.fasta --query in/query.fasta --outputdir out/
```

This will create two files in the `output` directory:

- `ALLELES.aln` : tab-delimited text file describing alignment of sequence reads to contig(s) in `ALLELES.fasta`
- `ALLELES.fasta` : fasta file of the largest constructed contig (allele) containing the initial query

- `ALLELES.tsv` the with the following columns (see example further below):

- `sseqid`  
\* name of sequencing read (from `reads.fastq`)
- `qseqid`  
\* name of contig matched (from `ALLELES.fasta`)
- `sstart`  
\* starting coordinate in sequencing read `sseqid` that matches `qseq`
- `send`  
\* ending coordinate in sequencing read `sseqid` that matches `qseq`
- `qstart`  
\* starting coordinate in contig that matches `sseq`
- `qend`  
\* ending coordinate in contig that matches `sseq`

Example `output.aln` file:

| <code>sseqid</code> | <code>qseqid</code> | <code>sstart</code> | <code>send</code> | <code>qstart</code> | <code>qend</code> |
|---------------------|---------------------|---------------------|-------------------|---------------------|-------------------|
| 2S43D:08461:04180   | contig1             | 13                  | 40                | 1                   | 64                |
| 2S43D:07701:07310   | contig1             | 20                  | 112               | 240                 | 332               |
| 2S43D:07489:10315   | contig1             | 123                 | 90                | 20                  | 53                |
| 2S43D:04035:14719   | contig1             | 105                 | 41                | 10                  | 74                |